

NDN Applications

NDNComm 2014 – ICN Tutorial Dry Run

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NDN: Application-motivated Approach

NSF FIA 2010-present

Video streaming, live chat, file sharing, lighting control, sensing



NSF FIA “Next Phase” (2014-2016)

Enterprise Building Automation & Management

Open Mobile Health

Mobile Multimedia Applications



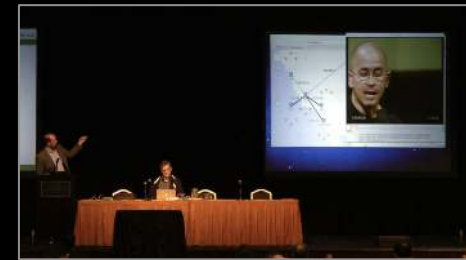
Other ongoing work (2014-2016)

Internet of Things, Raspberry PI support

Climate Modeling (Colorado State)

Information Maximizing Networks (UIUC)

Vehicular Networking (UPMC, UCLA)



ChronoChat

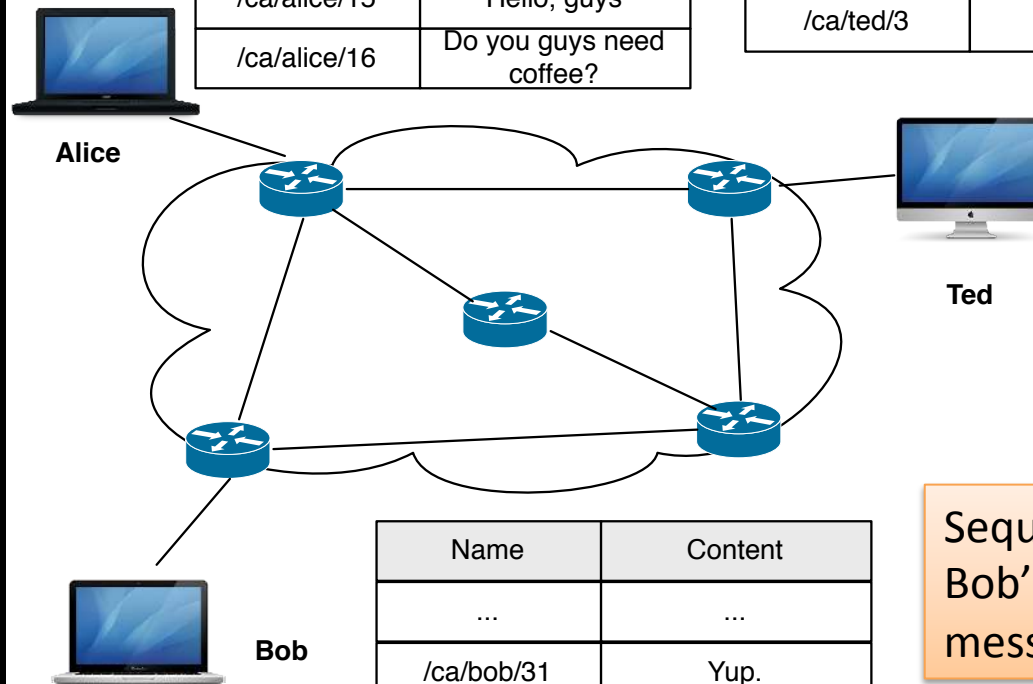
What is a group text chat application?

Sequence of
Alice's
messages

Name	Content
...	...
/ca/alice/15	Hello, guys
/ca/alice/16	Do you guys need coffee?

Sequence of
Ted's
messages

Name	Content
...	...
/ca/ted/3	Hello, Alice.



Name	Content
...	...
/ca/bob/31	Yup.

Sequence of
Bob's
messages

Synchronization of distributed chat room dataset (set of sequences of chat messages) among the participants

ChronoChat

Many Internet applications are collaborative by nature

- group text chat
- file sharing
- audio/video conferencing

Key piece in these applications

- distributed state synchronization
 - chat room messages
 - files and folders in the shared folder
 - voice/video streams from each participant

Based on ChronoSync

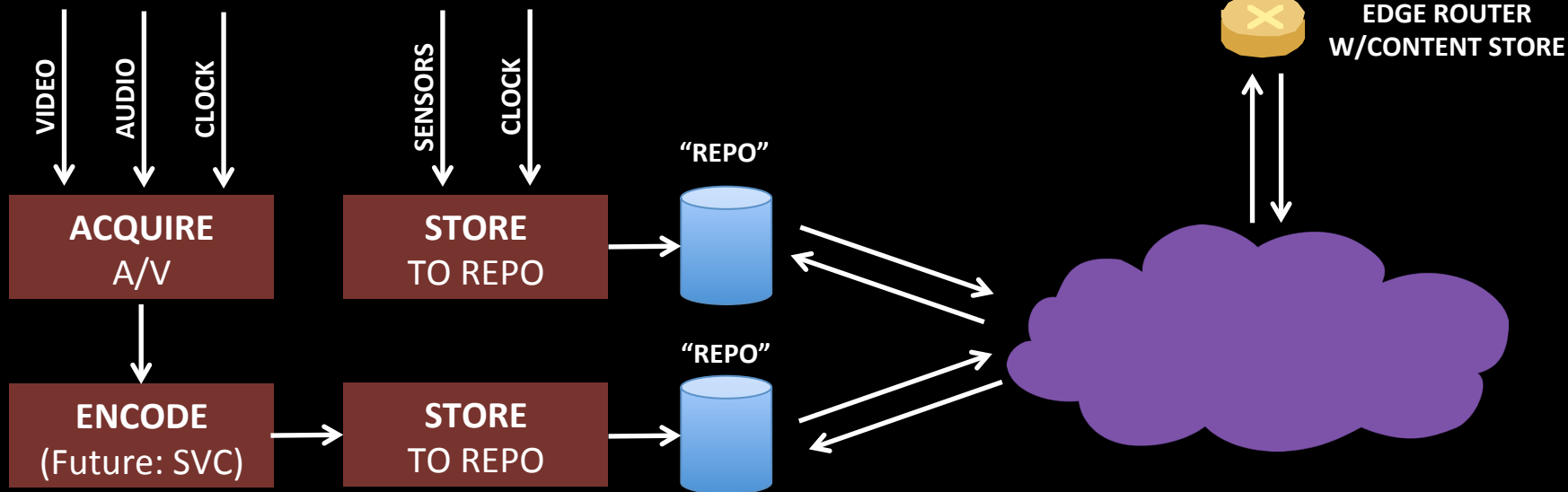
- synchronize the knowledge of the chat message set

Two trust models have been experimented

- hierarchical
 - authenticate users through a certificate chain that strictly follow the naming hierarchy
- web-of-trust
 - authenticate users through endorsements made by each other's directly/indirectly trusted users

NDNVideo

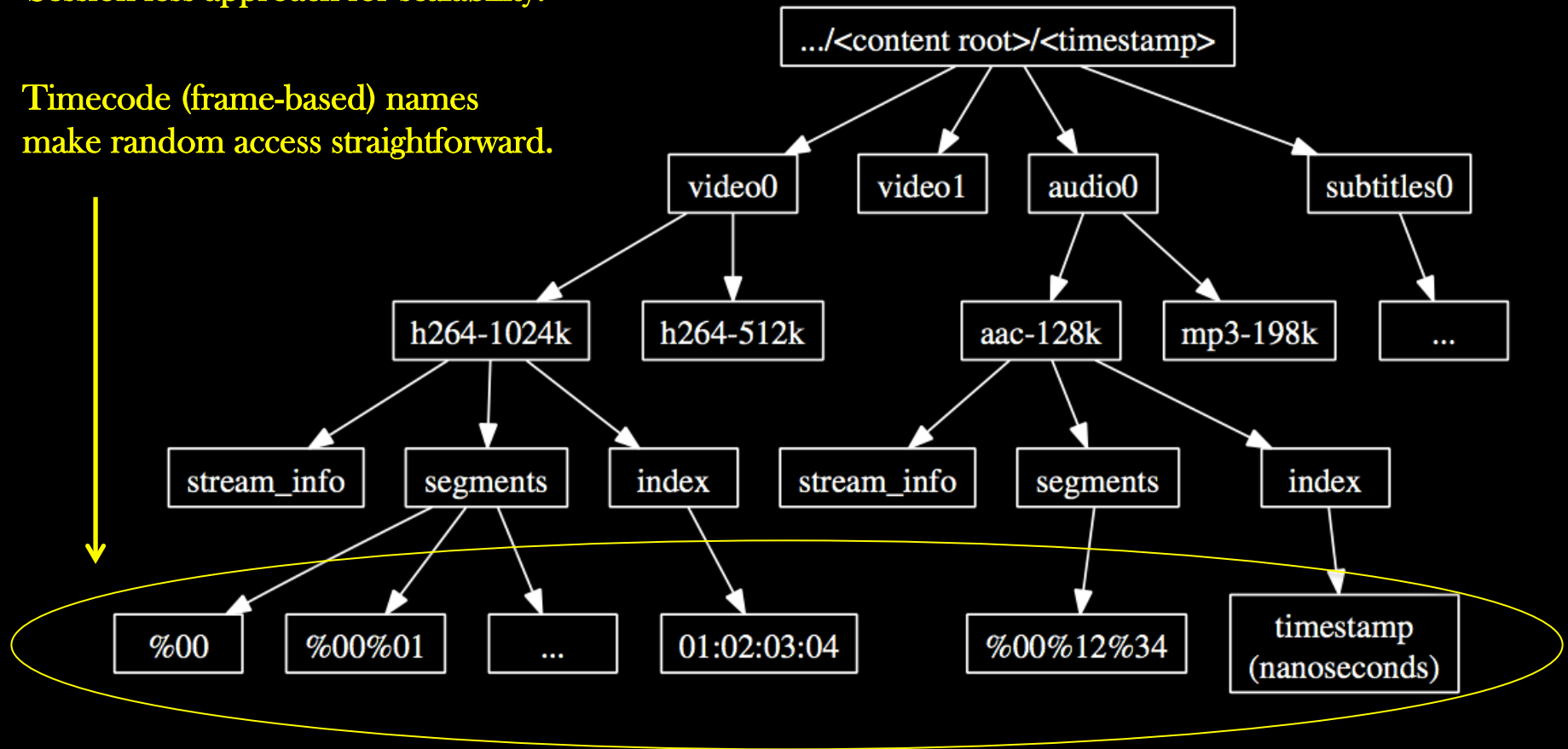
NDNVideo: Sessionless, Scalable Live, Pre-recorded Streaming



Design App Namespace to Leverage Architecture

Session-less approach for scalability.

Timecode (frame-based) names make random access straightforward.

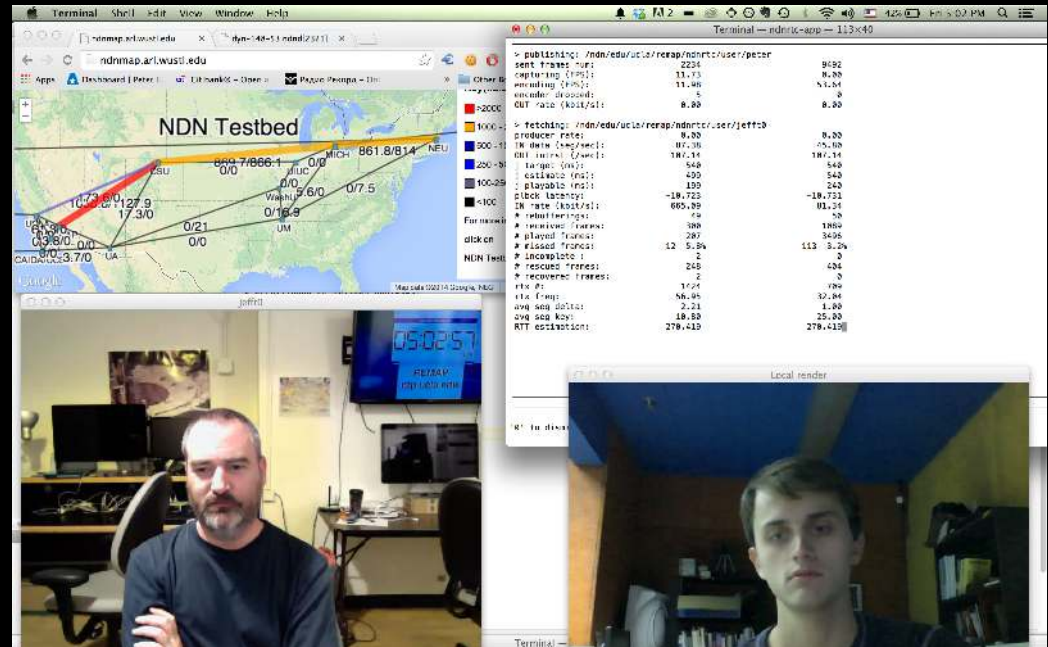
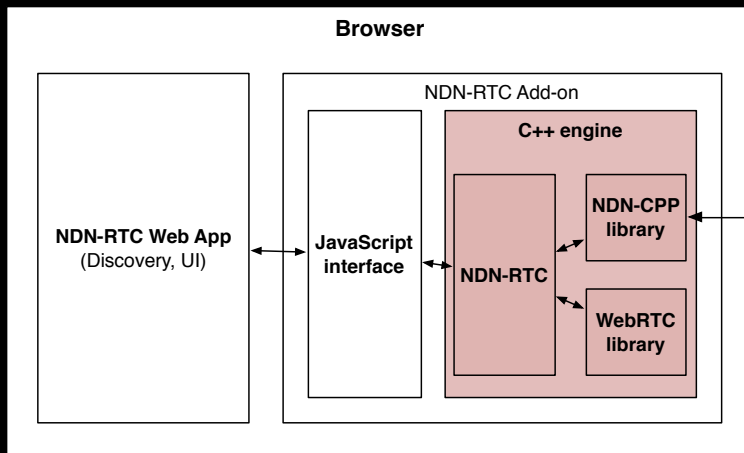


ndnrtc

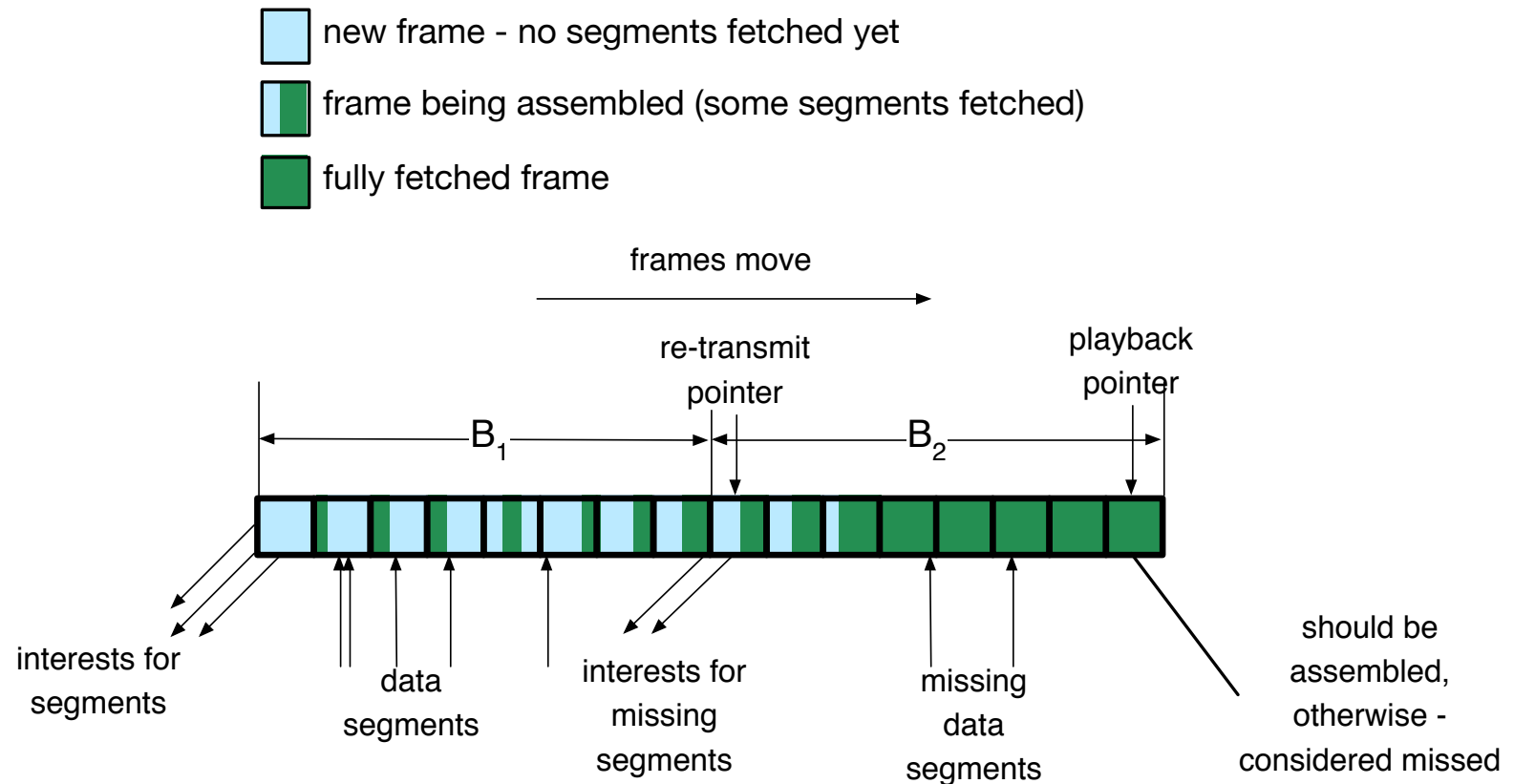
NDN Real Time Conferencing Tool

Goals:

- Real-time audio/video/text chat library which allows many-to-many conferencing over the NDN network and requires no direct communication between peers
- Traffic generator for the testbed
- Start point for NDN traffic congestion control algorithm research
- Test NDN-CPP library and TLV packet format



Deadline-based pipeline for driving interest retransmission



$B_1 \geq \text{RTT}, B_2 \geq \text{RTT}$

Minimal buffer size $\geq 2 \cdot \text{RTT}$ milliseconds

Vehicular networking

Applying NDN to vehicle networking

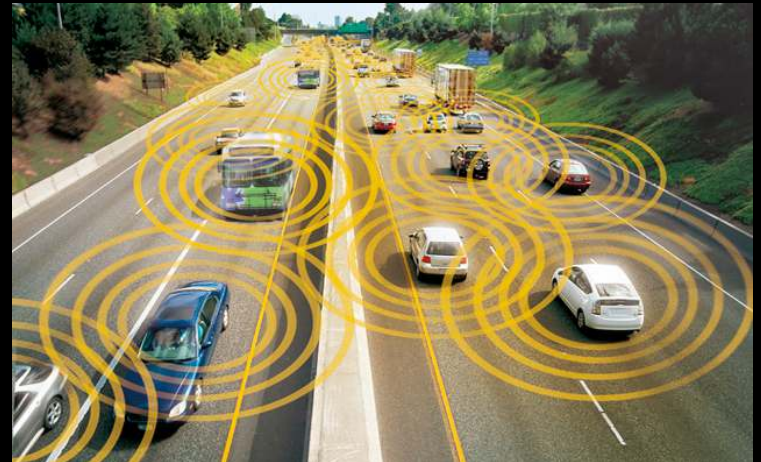
Question via the network: *Collisions ahead on I-80?*

```
/toyota/collision/by_road/us/california/I-80  
/toyota/collision/recent
```

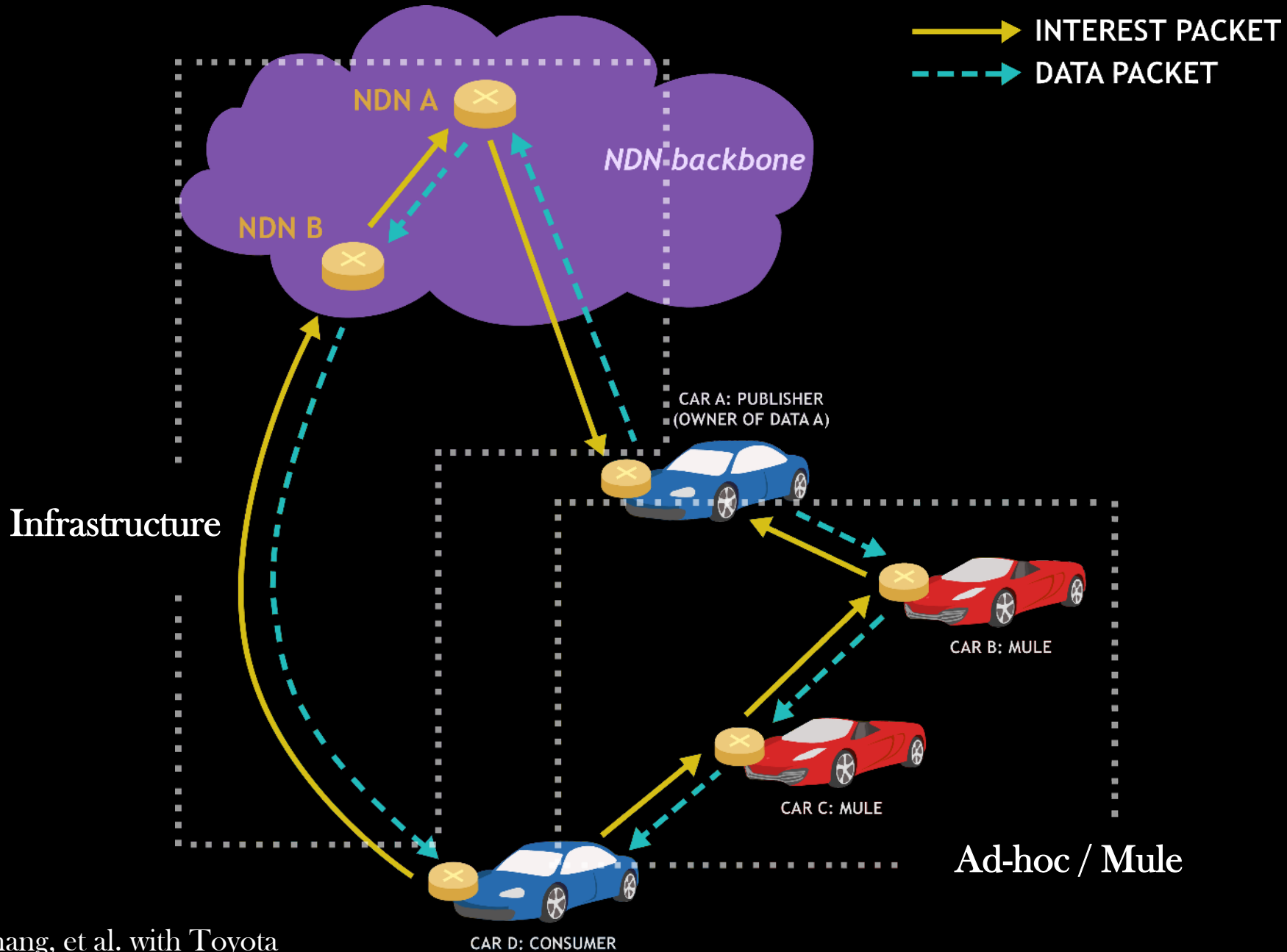
Research goal: a *single framework* for vehicles

- to fully utilize all available physical channels
- to communicate in an infrastructure-free manner
- to communicate with infrastructure servers
- to provide delay-tolerant delivery

*No need to deal with host addresses;
still must design application namespace.*



(Mobile) data gets to (mobile) consumers any way it can



Climate Modeling

NDN Approach



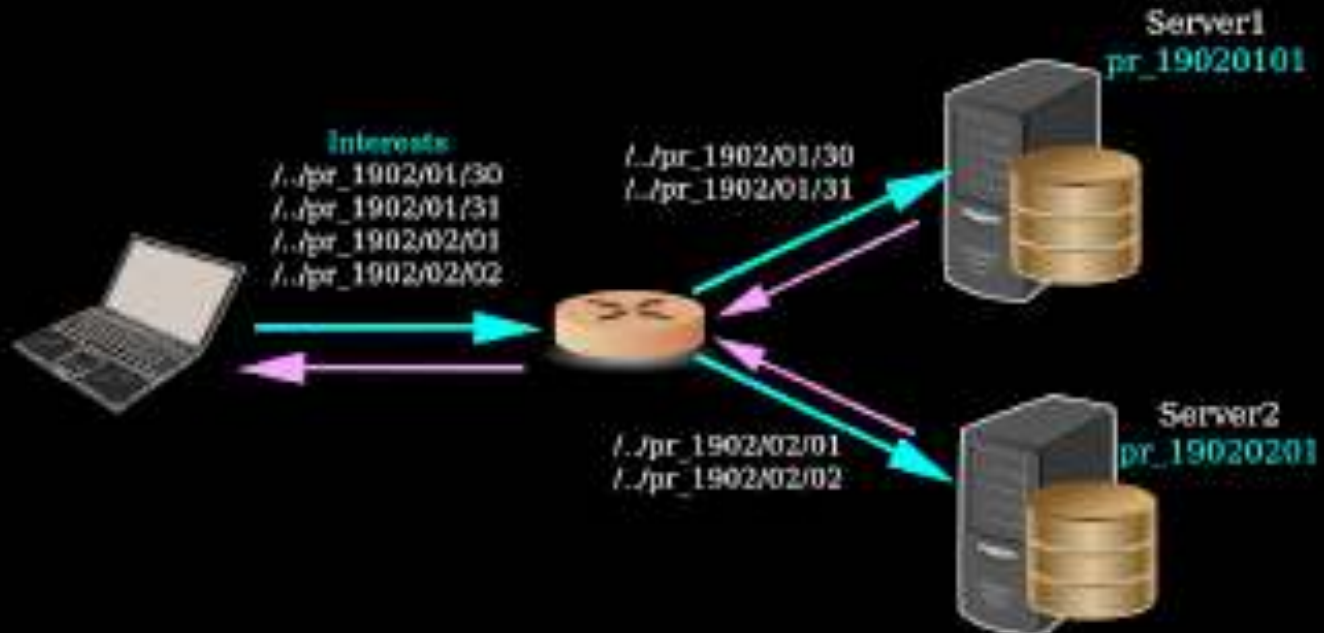
Consumer Request

- Client wants Jan 30 – Feb 02
- NDN routes Jan requests to Server1 and Feb requests to Server2

```
/cmmmap/precipitation/  
  GCRM/GridZ/  
    <horiz_resolution>/  
    <field>/<date>/<time>/
```

Publisher Announcement

- Server 1 advertises January prefix
- Server2 advertises February prefix
- Routing protocol propagates announcements
- Servers answer at appropriate granularity for the application



Naming in Climate Modeling Application

CMIP5: “.. a common naming system to be used in files, directories, metadata, and URLs to identify datasets wherever they might be located within the distributed CMIP5 archive. It defines controlled vocabularies for many of the components comprising the data reference syntax (DRS).

Directory encoding:

- /<activity>/<product>/<institute>/<model>/<experiment>/ <frequency>/<modeling realm>/ <variable name>/<ensemble member>
- **Example:** /CMIP5/output/MOHC/HadCM3/decadal1990/day/atmos/tas/r3i2p1/

CMIP5 DRS is NDN-compliant

Hierarchical

Clearly defined name components

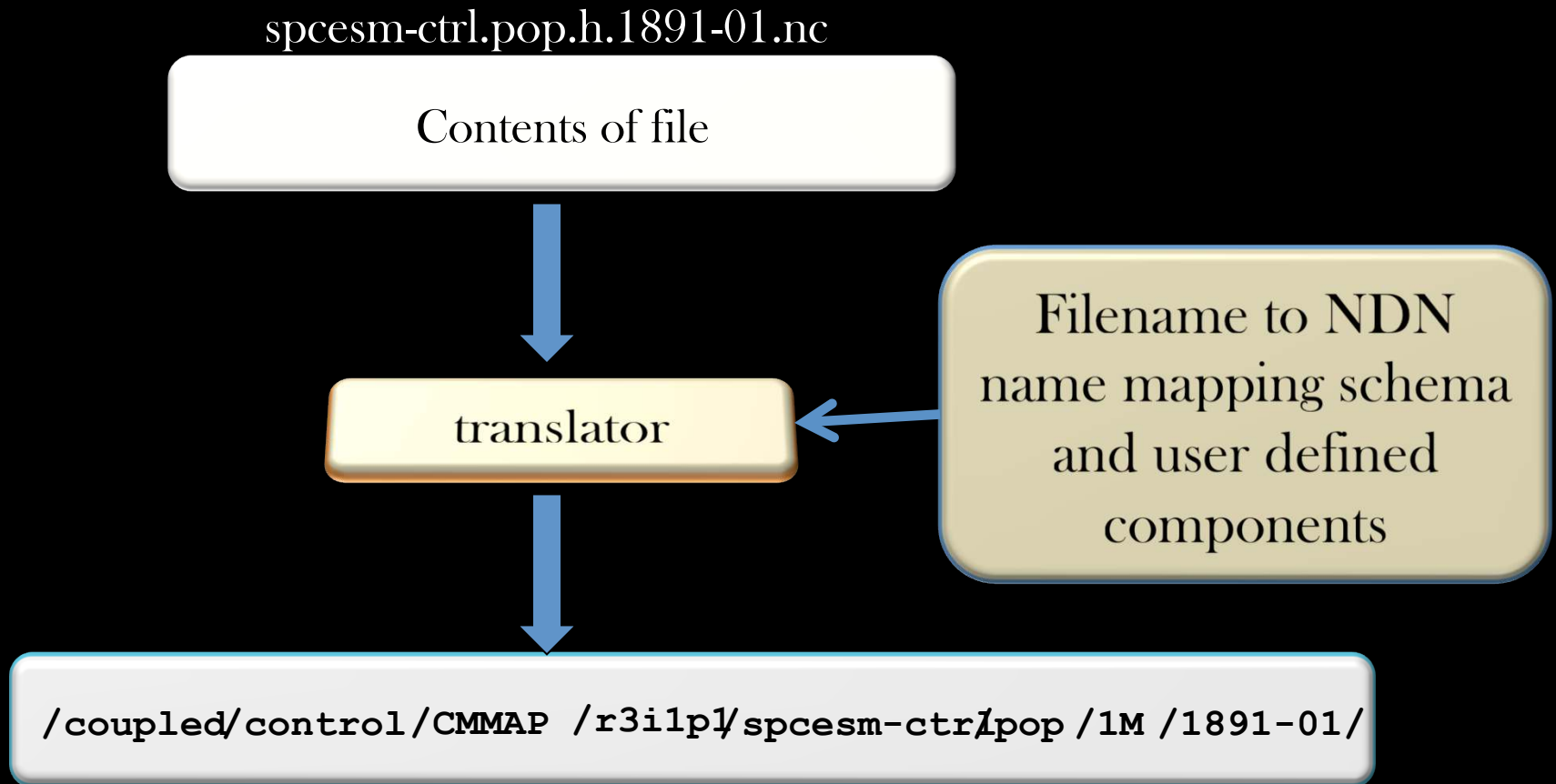
Well-defined vocabularies

Organizes components from less specific to more specific

Utilities to translate into DRS (CMOR)

Produces virtually ready-to-use NDN names

Our Translator Architecture



activity/sub-activity/organization/ensemble/experiment/model/
granularity/start-time

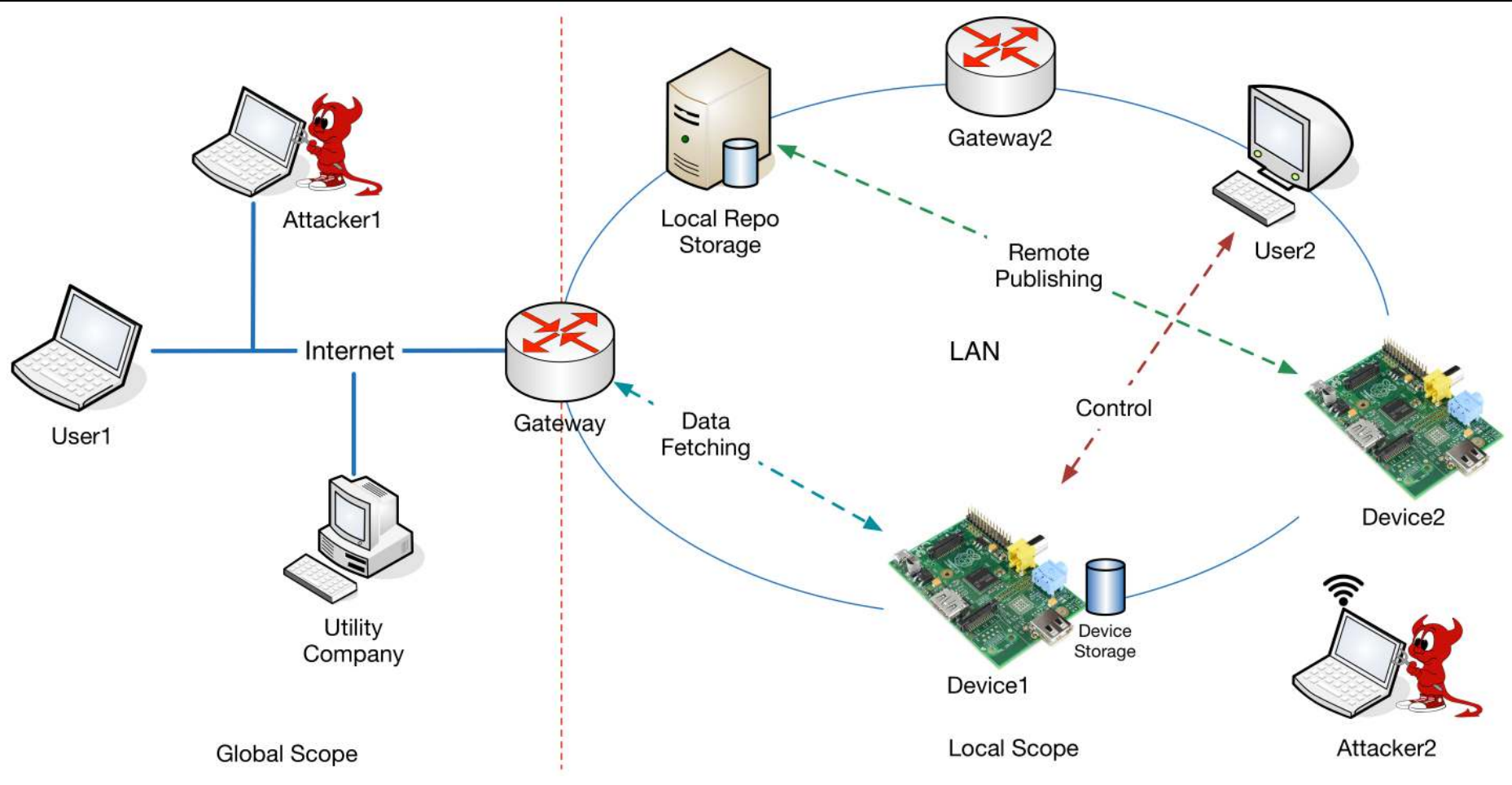
NP Network Environments

Enterprise building automation
and management systems

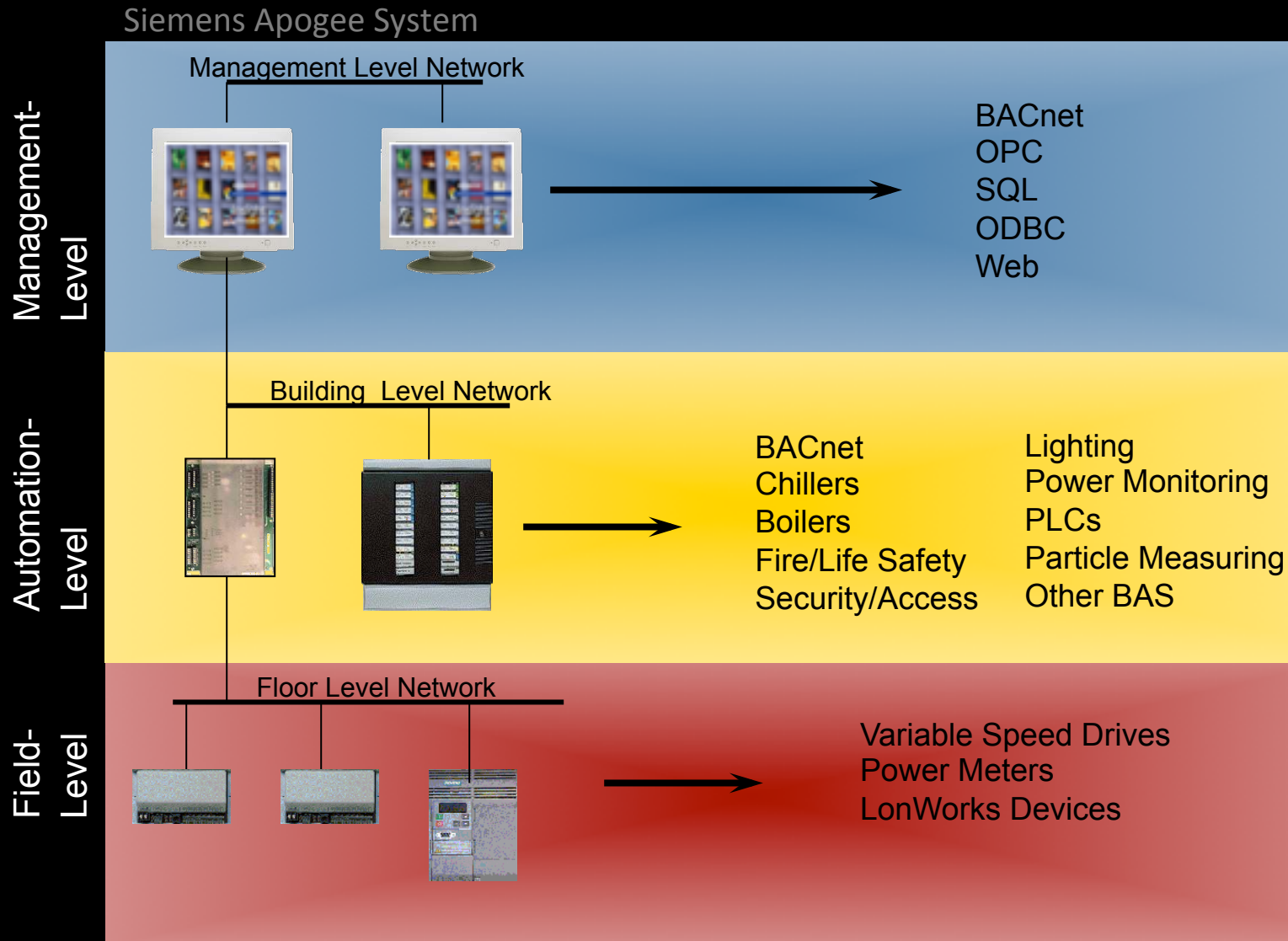
Two research threads

- Enterprise-level BAS/BMS in collaboration with UCLA Facilities Management, based on work started in an EAGER.
- Device-side IoT, motivated by consumer experience / home environment for now. (External support from Qualcomm and potentially Huawei.)

Home environment



EBAMS: Focusing on “Management” / “Automation” levels



Sensing: Electrical Demand & Chilled Water



Two testbeds at UCLA – one shared and one unique.



NDN – Suitability / Benefits

Massive addressing simplification, with a potential for huge impact when scaled to the enterprise. Simpler network infrastructure needed to deploy complex monitoring and automation.

New way of working with edge resources that de-emphasizes gateway addressing while preserving support for topological heterogeneity.

Lighter-weight, data-centric security options easier to develop, with data verification intrinsically part of the architecture.

Caching and storage integration may provide significant advantages in distributed storage at all levels of the architecture, increasing data availability without power increase.

Intrinsic multicast; many-to-many communication easier to deploy.

Hierarchical naming already used at application layer

MLNTZ.PNL.J.SAT	LAO	SAT TOTAL	Value	0.0.0	No	No
MLNTZ.PNL.J.SEP	LAO	SEP TOTAL	Value	0.0.0	No	No
MLNTZ.PNL.J.SUN	LAO	SUN TOTAL	Value	0.0.0	No	No
MLNTZ.PNL.J.THU	LAO	THU TOTAL	Value	0.0.0	No	No
MLNTZ.PNL.J.TUE	LAO	TUE TOTAL	Value	0.0.0	No	No
MLNTZ.PNL.J.WED	LAO	WED TOTAL	Value	0.0.0	No	No
MLNTZ.PNL.J.WEK	LAO	WEEK TOTAL	Value	0.0.0	No	No
MLNTZ.PNL.J.YPEAK	LAO	YESTERDAYS PEAK	Value	0.0.0	No	No
MLNTZ.PNL.J.CNSMTN.LO	LAI	ACTUAL LO CNSMTN	Value	1.1.8	No	No
MLNTZ.PNL.J.CNSMTN.HI	LAI	ACTUAL HI CNSMTN	Value	1.1.9	No	No
MLNTZ.PNL.J.DEMAND	LAI	ACTUAL VOLTS	Value	1.1.10	No	No
MLNTZ.PNL.J.AMPS	LAI	ACTUAL AMPS	Value	1.1.12	No	No
MLNTZ.PNL.J.VOLTS	LAI	ACTUAL VOLTS	Value	1.1.13	No	No
MLNTZ.PNL.J.DEM:DAY.NGT	LDO	300 AMP	On/Off	1.1.29	No	No

current
volts

through hole
v battery

inherent value

UCLA NDN Building Monitoring Testbed

Snapshot - Strathmore Snapshot - Melnitz All data - Melnitz About

Melnitz BACnet data summary:

BACnet Data Point	Timestamp	Value	Unit
MLNTZ.STUDIO1.DEMAND	23:30:45	5.9039	kW
MLNTZ.STUDIO1.PEAK	23:30:46	19.487	kW
MLNTZ.STUDIO1.A405	23:30:46	0.6184	kW
MLNTZ.STUDIO1.A410	23:30:47	1.2296	kW
MLNTZ.STUDIO1.A415	23:30:47	1.8407	kW
MLNTZ.STUDIO1.A4DC	23:30:48	4	kW
MLNTZ.STUDIO1.C7	23:30:49	1499.6	kW
MLNTZ.STUDIO1.C7AVG	23:30:49	214.24	kW
MLNTZ.STUDIO1.MON	23:30:50	192.08	kW
MLNTZ.STUDIO1.VOLTS	23:30:50	213.09	V
MLNTZ.STUDIO1.AMPS	23:30:51	20.875	A
MLNTZ.PNL.DMR.DEMAND	23:30:52	0	kW
MLNTZ.PNL.DMR.PEAK	23:30:52	10.496	kW
MLNTZ.PNL.DMR.VOLTS	23:30:53	213	V
MLNTZ.PNL.DMR.AMPS	23:30:53	0	A
MLNTZ.PNL.AHB.DEMAND	23:30:54	5.7919	kW
MLNTZ.PNL.AHB.PEAK	23:30:55	5.8400	kW
MLNTZ.PNL.AHB.VOLTS	23:30:55	212.5	V
MLNTZ.PNL.AHB.AMPS	23:30:56	19.125	A
MLNTZ.PNL.AA.DEMAND	23:30:58	0	kW
MLNTZ.PNL.AA.PEAK	23:30:59	0	kW
MLNTZ.PNL.AA.VOLTS	23:30:59	213	V

Powered by NDNJS

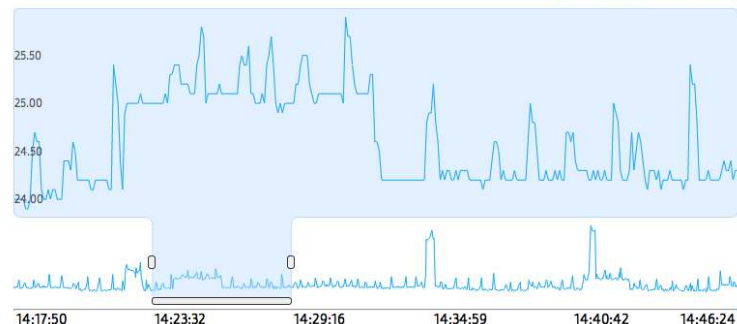
UCLA NDN Building Monitoring Testbed

Snapshot - Strathmore Snapshot - Melnitz All data - Melnitz About

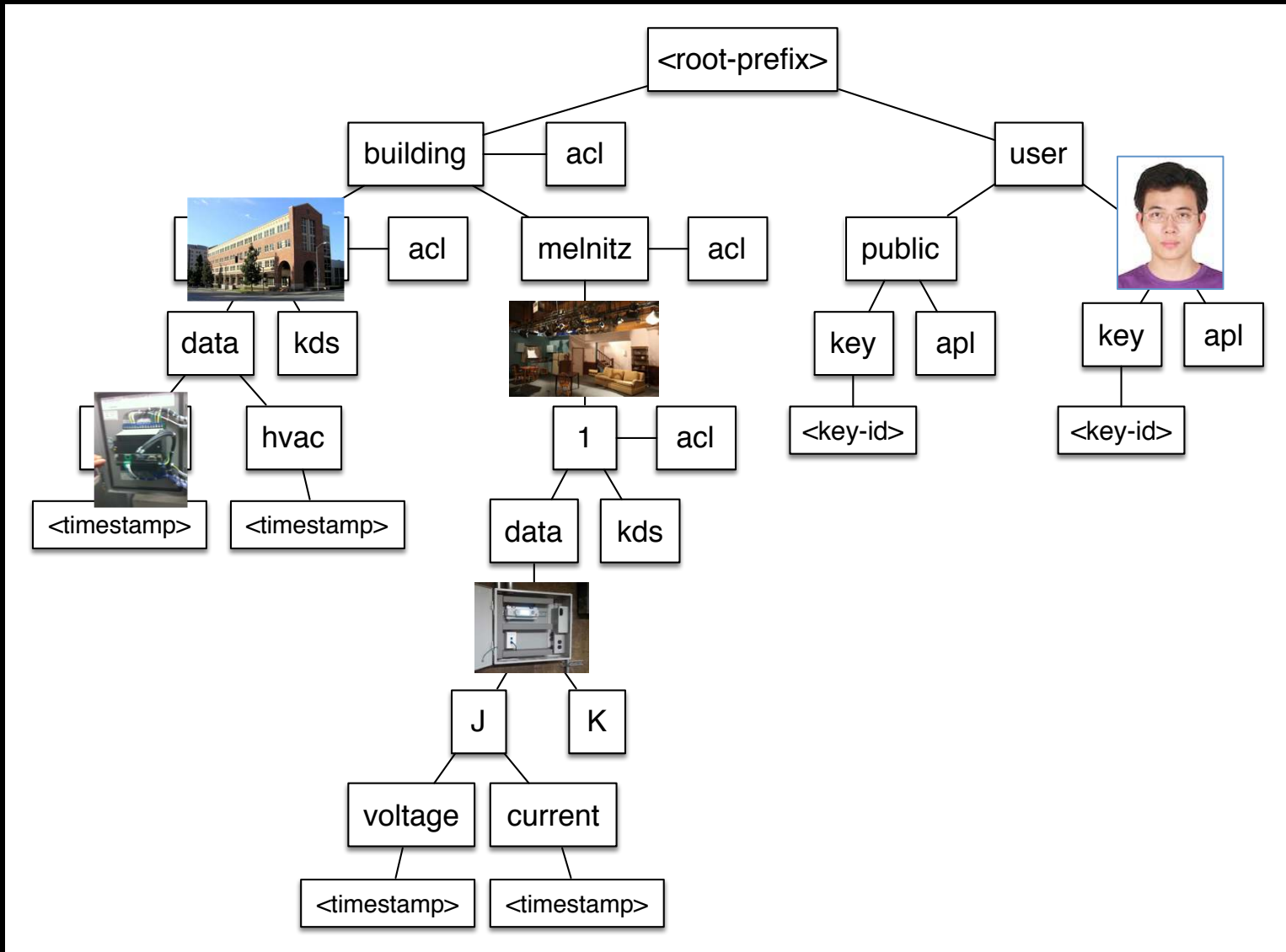
Strathmore Building



Electrical Demand - Current (unit: Amperes)



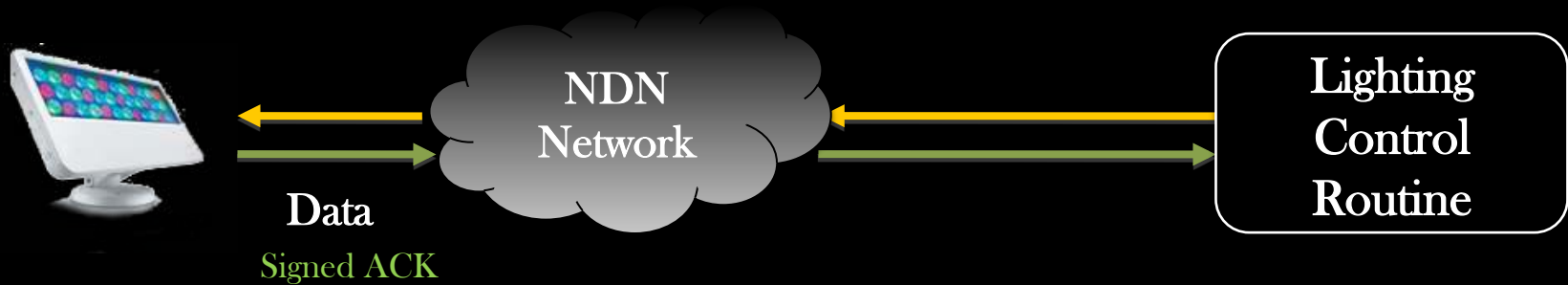
Forward based on names closely mapped to physical world



Authenticated, Closed-Loop Control in NDN

Interest **signed by app**

boelter/3551/lights/fixture/41/rgb-8bit-hex/FAF87F/<state>/<authenticator>



- Asymmetric keys to work directly with PK-based identities
- Symmetric keys and HMACs for faster 'signatures'
- Leverage NDN to distribute keys and establish key relationships
- Command privacy by encrypting non-routable portion of name

Open mHealth

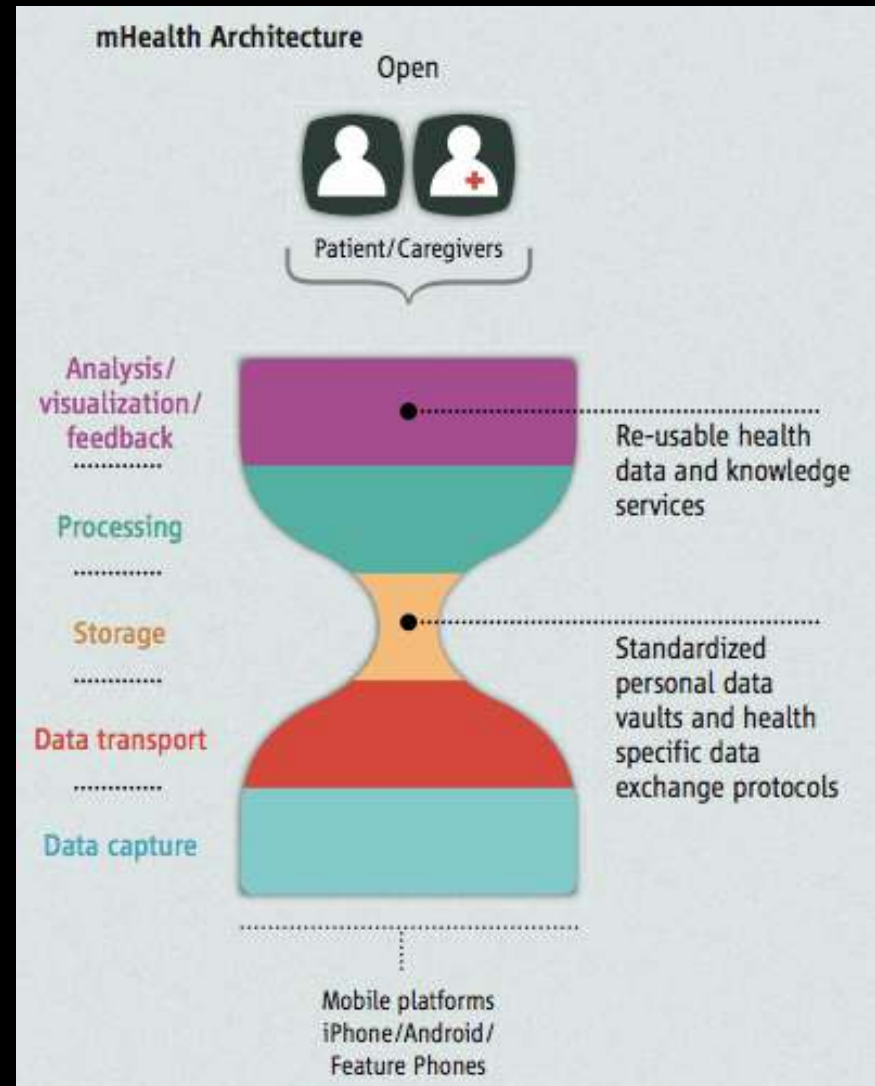
Open mHealth: Data rather than System Interoperability

Interoperable, Internet-inspired data exchange as the backbone of the application ecosystem

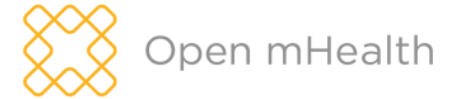
Thin waist of open data interchange standards that will enable an ecosystem of sensing, storage, analysis, and user interface components to support medical discovery and evidence-based care

Market-supported, patient-centered landscape of innovative health applications

Patient-controlled, privacy-aware data exchange across device, component, and application boundaries



Mapping the Open mHealth Architecture to NDN



Data Visualization Units embody the semantics necessary to meaningfully **display** a representation of data.

registry.openmhealth.org

Composability &
Authentication

Data Processing Units embody the semantics necessary to meaningfully **process** a representation of data. E.g., convert a day of steps into the diameter of a day (convert one representation into another).

Access Audit

Storage Design

Request / Response model

All Open mHealth Data Storage Units are required to implement a **small HTTP-based specification** for data access.

Provenance

Namespace design

The Open mHealth **schemas** present in the data store.

Same Challenges, Different Layers

For this application in particular, NDN provides much more relevant functionality at the network layer than IP.

So solutions in NDN have much more direct impact on the scalability, security, and ease of development; we need not build up additional layers on IP to get near the app challenges.

Namespace / schema design

Repository / storage design

Service composability

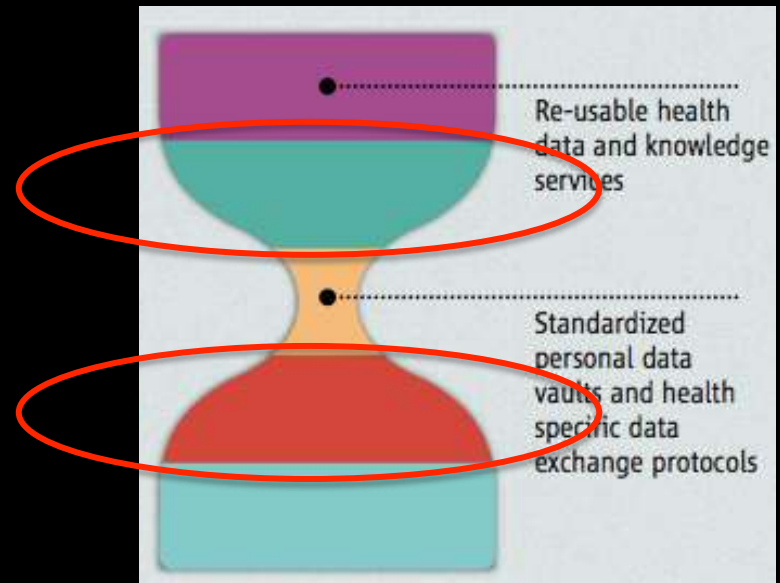
Authentication / identity assurance

Data provenance

Access auditing

Mobile publishing

Legal requirements for success



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